Dying Alone

An interview with Eric Klinenberg

author of *Heat Wave: A Social Autopsy of Disaster in Chicago*

Question: Take us back to July 1995 in the city of Chicago. How hot was it? What were the city and its residents going through?

Klinenberg: Chicago felt tropical, like Fiji or Guam but with an added layer of polluted city air trapping the heat. On the first day of the heat wave, Thursday, July 13, the temperature hit 106 degrees, and the heat index—a combination of heat and humidity that measures the temperature a typical person would feel—rose above 120. For a week, the heat persisted, running between the 90s and low 100s. The night temperatures, in the low to mid-80s, were unusually high and didn't provide much relief. Chicago's houses and apartment buildings baked like ovens. Air-conditioning helped, of course, if you were fortunate enough to have it. But many people only had fans and open windows, which just recirculated the hot air.

The city set new records for energy use, which then led to the failure of some power grids—at one point, 49,000 households had no electricity. Many Chicagoans swarmed the city's beaches, but others took to the fire hydrants. More than 3,000 hydrants around Chicago were opened, causing some neighborhoods to lose water pressure on top of losing electricity. When emergency crews came to seal the hydrants, some people threw bricks and rocks to keep them away.

COVER
TO
COME

The heat made the city's roads buckle. Train rails warped, causing long commuter and freight delays. City workers watered bridges to prevent them from locking when the plates expanded. Children riding in school buses became so dehydrated and nauseous that they had to be hosed down by the Fire Department. Hundreds of young people were hospitalized with heat-related illnesses. But the elderly, and especially the elderly who lived alone, were most vulnerable to the heat wave.

After about forty-eight hours of continuous exposure to heat, the body's defenses begin to fail. So by Friday, July 14, thousands of Chicagoans had developed severe heat-related illnesses. Paramedics couldn't keep up with emergency calls, and city hospitals were overwhelmed. Twenty-three hospitals—most on the South and Southwest Sides—went on bypass status, closing the doors of their emergency rooms to new patients. Some ambulance crews drove around the city for miles looking for an open bed.

Hundreds of victims never made it to a hospital. The most overcrowded place in the city was the Cook County Medical Examiners Office, where police transported hundreds of bodies for autopsies. The morgue typically receives about 17 bodies a day and has a total of 222 bays. By Saturday—just three days into the heat wave—its capacity was exceeded by hundreds, and the county had to bring in a fleet of refrigerated trucks to store the bodies. Police officers had to wait as long as three hours for a worker to receive the body. It was gruesome and incredible for this to be happening in the middle of a modern American city.

Question: How many people died as a result of the heat wave?

Klinenberg: In 1995 there were no uniform standards for determining a "heat related death," so officials had to develop them. Edmund Donoghue, Cook County's chief medical examiner, used state-of-the-art criteria to report 465 heat-related deaths for the heat wave week and 521 heat deaths for the month of July. But Mayor Richard M. Daley challenged these findings. "It's hot," the mayor told the media. "But let's not blow it out of proportion.... Every day people die of natural causes. You cannot claim that everybody who has died in the last eight or nine days dies of heat. Then everybody in the summer that dies will die of heat." Many local journalists shared Daley's skepticism, and before long the city was mired in a callous debate over whether the so-called heat deaths were—to use the term that recurred at the time—"really real."

Medical examiners around the country confirmed that Donoghue's heat-related death criteria were scientifically sound and endorsed his findings. But perhaps the best measure of heat deaths comes from another figure—the "excess death" rate—which counts the difference between the reported deaths and the typical deaths for a given time period. According to this measure, 739 Chicagoans *above the norm* died during the week of 14 to 20 July—which means that Donoghue had been conservative in his accounts.

Daley's skepticism had a big impact on the public debate, and it still does. Today if you ask Chicagoans about the heat wave they will likely tell you that not all the deaths were "really real." That's a direct legacy of the politics of the disaster.

Question: Who were these 739 people? Was there a "typical" victim?

Klinenberg: The US Centers for Disease Control and Prevention did a thorough study of individual-level risk factors for heat wave victims, and they came up with a list of conditions of vulnerability: living alone, not leaving home daily, lacking access to transportation, being sick or bedridden, not having social contacts nearby, and of course not having an air conditioner.

Given these factors, experts assumed that female victims would outnumber male victims in the heat wave deaths, because women are more prevalent among those who are old and who live alone. But in fact men were more than twice as likely to die as women. This is just one of the surprises that emerged during my study of the Chicago heat wave. To understand this we have to look at the social relationships that elderly women retain but that elderly men tend to lose.

The ethnic and racial differences in mortality are also significant for what they can teach us about urban life. The actual death tolls for African Americans and whites were almost identical, but those numbers are misleading. There are far more elderly whites than elderly African Americans in Chicago, and when the Chicago Public Health Department considered the age differences, they found that the black/white mortality ratio was 1.5 to 1.

Another surprising fact that emerged is that Latinos, who represent about 25 percent of the city population and are disproportionately poor and sick, accounted for only 2 percent of the heat-related deaths. I wrote *Heat Wave* to make sense of these numbers—to show, for instance, why the Latino Little Village neighborhood had a much lower death rate than African American North Lawndale. Many Chicagoans attributed the disparate death patterns to the ethnic differences among blacks, Latinos, and whites—and local experts made much of the purported Latino "family values." But there's a social and spatial context that makes close family ties possible. Chicago's Latinos tend to live in neighborhoods with high population density, busy commercial life in the streets, and vibrant public spaces. Most of the African American neighborhoods with high heat wave death rates had been abandoned—by employers, stores, and residents—in recent decades. The social ecology of abandonment, dispersion, and decay makes systems of social support exceedingly difficult to sustain.

Question: So would you call the heat wave deaths primarily a social disaster, rather than a natural one?

Klinenberg: Of course forces of nature played a major role. But these deaths were not an act of God. The authors of an article in the *American Journal of Public Health* said that the most sophisticated climate models "failed to detect relationships between the weather and mortality that would explain what happened in July 1995 in Chicago." Hundreds of Chicago residents *died alone,* behind locked doors and sealed windows, out of contact with friends, family, and neighbors, unassisted by public agencies or community groups. There's nothing natural about that.

The death toll was the result of distinct dangers in Chicago's social environment: an increased population of isolated seniors who live and die alone; the culture of fear that makes city dwellers reluctant to trust their neighbors or, sometimes, even leave their houses; the abandonment of neighborhoods by businesses, service providers, and most residents, leaving only the most precarious behind; and the isolation and insecurity of single room occupancy dwellings and other last-ditch low-income housing. None of these common urban conditions show up as causes of death in the medical autopsies or political reports that establish the official record for the heat disaster.

Chicago had such a high mortality rate because it is, as Mayor Daley quipped during the heat wave, the classic American city of extremes. It is a city of great opulence and of boundless optimism, but—as William Julius Wilson says—Chicago also suffers from an everyday "emergency in slow motion" that its leaders refuse to acknowledge. The heat wave was a particle accelerator for the city: It sped up and made visible the hazardous social conditions that are always present but difficult to perceive. Yes, the weather was extreme. But the deep sources of the tragedy were the everyday disasters that the city tolerates, takes for granted, or has officially forgotten.

end

Question: What about the response from the city? Did the city government do enough to warn residents of the danger, provide cool shelter, or help people who were in trouble? What could Chicago do differently in future heat waves?

Klinenberg: It is not fair to blame any single organization or individual for an event in which hundreds of people die alone. The heat disaster was a collective failure, and the search for scapegoats—whether the mayor, the media, or the medical system—is just a distraction from the real issues.

Yet there is no question that the city government did not do everything it could to prevent the catastrophe. The city failed to implement its own heat emergency plan, waiting until Saturday, July 15, after hundreds of bodies had already been delivered to the county morgue, to declare an official emergency. The Fire Department refused its paramedics' requests to call in more staff and secure more ambulances, thereby assuring continued delays in its emergency health response. The Police Department did not use its senior units to attend to the elderly residents they were supposed to protect. And since there was no system to monitor the hospital bypass situation, at one point eighteen hospitals were simultaneously refusing new emergency patients.

The city also aggressively used its tremendous public relations apparatus to first deny there was a disaster and then to define the disaster as natural and unpreventable. The might seem bizarre. He explained it by way of a story: Sometimes, he said, he would pick up his three-year-old son from day care and put him on the backseat of his tandem bike and they would pedal home along the South Saskatchewan River. The snow would muffle the noise of the city. Dusk would paint the sky in colors so exquisite that Judge could not begin to find names for them. The snow would reflect those hues. It would glow like the sky, and Judge would breathe in the cold air and hear his son breathing behind him, and he would feel as though together they had become part of winter itself.

Few people share Judge's tolerance for discomfort, hard work, and inconvenience, but most of us are more like him than we might imagine. Our urban journeys can meet all kinds of psychological needs. "For many, the commute really is a kind of heroic quest," Patricia Mokhtarian, a University of California, Davis, transportation engineer, said after I told her Judge's story. She said many car commuters feel the same way. "Remember the *Odyssey*, where the heroes launch their ships and head off to face adventures and traumas before making their return? Well, the commute can be this heroic going out into the world, conquering the traffic, surviving, and coming home to the warm reception of family."

People may complain about commuting, but after surveying hundreds of commuters in California, Mokhtarian discovered that the average person actually prefers to be forced to travel for part of every day. "We hear many people say, 'Darn, my commute is not long enough!'" Of course, few people pine for a super-commute. The trip time most people *wish* they had is about sixteen minutes, one way.* Still, Mokhtarian and other travel researchers insist that long or short, every commute is a ritual that can alter our very sense of who we are and what is our place in the world.

*Like the people in her studies, Mokhtarian likes the ritual transition between home and work. In fact, rather than living near her office in the cozy hamlet of Davis, she chose to live in the nearby town of Woodland, a conscious choice that forced her to drive to work each day. Commute time? Sixteen minutes, door-to-door.

Driving Sideways

If you were to judge the hedonic utility of various modes of travel by how many people choose them each day, there would be absolutely no substitute for driving an automobile—at least not in North America. Nearly nine in ten American commuters drive to work every day. Three-quarters of Canadians and two-thirds of Brits do the same.

Drivers experience plenty of emotional dividends. When the road is clear, driving your own car embodies the psychological state known as mastery: drivers report feeling much more in charge of their lives than transit users or even their own passengers. Many commuters admitted to Mokhtarian that much of the pleasure of driving came simply from being seen in their fine cars. An upmarket vehicle is loaded with symbolic value that offers a powerful, if temporary, boost in status. The biochemical response is especially strong in young men. Researchers in Montreal found that when male college students spent a mere hour driving an expensive sports car—a $150,000 Porsche—they experienced a heady blast of testosterone, while driving an older, high-mileage Toyota Camry left them slightly drained. "The endocrinological response was substantial, irrespective of whether they had an audience or not," explained study coauthor Gad Saad, associate professor of marketing at Concordia University. In other words, the experience of driving a hot car triggered a hormonal response even when there were no hot babes to impress. No wonder four in ten Americans actually claim to *love* their cars.

Despite these romantic feelings, half of commuters living in big cities and suburbs claim to dislike the heroic journey they must make every day—an unhappy group made up mostly of drivers. Part of the problem is that cars fail to deliver the experience of freedom and speed that we all know they are capable of bestowing in a world of open roads. The urban system neutralizes their power. Luxury and sports cars might still offer their drivers a status bump, but the car's muscles cease to matter when it is surrounded by other cars.

Driving in traffic is harrowing for both brain and body. The blood of people who drive in cities is a high-test stew of stress hormones. The worse the traffic, the more your system is flooded with adrenaline and cortisol, the fight-or-flight juices that, in the short term, get your heart pumping faster, dilate your air passages, and help sharpen your alertness, but in the long term can make you ill. It can take as much as an hour to recover the ability to concentrate after a long urban commute. Researchers for Hewlett-Packard convinced volunteers in England to wear electrode caps during their commutes and found that whether they were driving or taking the train, peak-hour travelers suffered worse stress than fighter pilots or riot police facing mobs of angry protesters.[*]

If you have ever flown a spaceship through an asteroid belt or driven the Santa Ana Freeway from Anaheim to Los Angeles on a Friday evening, you will understand and have benefited from the heightened focus and alertness offered by the full-on adrenal rush. It can be thrilling in the short term, but if you bathe in these hormones for too long, they can be toxic. Your immune system will be compromised, your blood vessels and bones will weaken, and your brain cells will begin to die off from the stress. Chronic road rage can actually alter the shape of the amygdalae, the brain's almond-shaped fear centers, and kill cells in the hippocampus.

This is part of the reason why urban bus drivers get sick more often, miss work more frequently, and die younger than people in other occupations. One stress-medicine specialist, Dr. John Larson, reported that many of his heart attack patients had one thing in common: shortly before their hearts gave out, they had been enraged while driving. No wonder people start to report steady drops in life satisfaction the more their commute time exceeds Mokhtarian's utopian sixteen minutes, even if they don't attribute their misery to their commute.[†]

[*]Commuters' hearts raced at 145 beats per minute, well over double the normal rate. They experienced a surge in cortisol. And, in what was apparently a coping strategy, their brains underwent a bizarre temporary transformation that psychologist David Lewis dubbed "commuter amnesia." Their brains simply shut out stimulus from the outer world, and they forgot about most of the trip as soon as it was over.

[†]When Gallup and Healthways polled Americans, they found that the longer people's commute, the more likely they were to report chronic pain, high cholesterol, and general unhappiness. (People with commutes over ninety minutes have it the worst. They are the most likely to be anxious, tired, and fat. And they are much less likely than people with short journeys to say they enjoy life.)

Cars once promised us unparalleled freedom and convenience, but despite fantastic investments in roads and highways, and the almost complete configuration of North American cities to favor automobile travel, commute times have been getting steadily longer. Americans, for example, clocked in relatively the same average daily commute times for years—about forty minutes round-trip, not including time spent on other errands—since as far back as 1800. But the average American now spends more than fifty minutes commuting. Return commute times have shot past sixty-eight minutes in the New York megalopolis, seventy-four minutes in London, and a whopping eighty minutes in Toronto. Dozens of studies have now confirmed beyond doubt what Atlantans know from experience: the obvious solution to congestion—building more roads—simply produces more traffic, creating a hedonic treadmill of construction and frustration.

Happy Feet

KIERKEGAARD

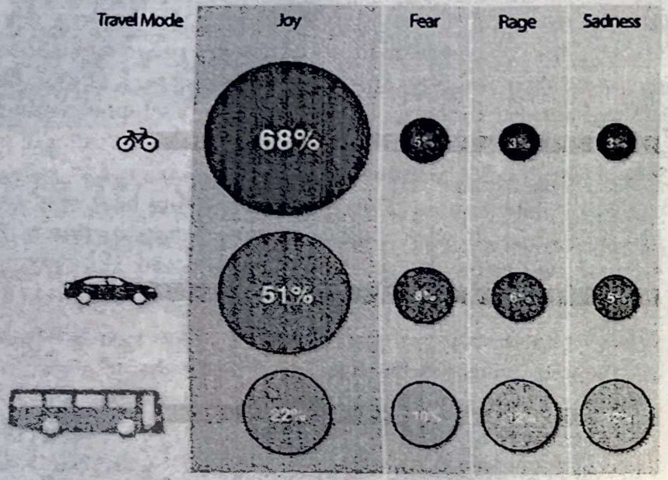
One group of commuters reports enjoying themselves more than everyone else. Their route to happy mobility is simple. These are people who travel on their own steam like Robert Judge. They walk. They run. They ride bicycles.

Despite the obvious effort involved, self-propelled commuters report feeling that their trips are *easier* than the trips of people who sit still for most of the journey. They are the likeliest to say their trip was fun. Children overwhelmingly say they prefer finding their own way to school rather than being chauffeured. These are the sentiments of people in American and Canadian cities, which tend to be designed in ways that make walking and cycling unpleasant and dangerous. In the Netherlands, where road designers create safe spaces for bikes, cyclists report feeling more joy, less fear, less anger, less sadness than both drivers and transit users. Even in New York City, where the streets are loud, congested, aggressive, and dangerous, cyclists report enjoying their journeys more than anyone else.

Why would traveling more slowly and using more effort offer more satisfaction than driving? Part of the answer exists in basic human physiology. We were born to move—not merely to be transported,

Happy Travels

Percentage of commuters reporting these emotions in the Netherlands:



In the Netherlands, where road space is provided for everyone, cyclists are by far the happiest people on the road. Public transit users report being the most miserable, as they do in most other places. (Scott Keck; from Harms, L., P. Jorritsma, and N. Kalfs, *Beleving en beeldvorming van mobiliteit*, The Hague: Kennisinstituut voor Mobiliteitsbeleid, 2007)

but to use our bodies to propel us across the landscape. Our geneti forebears have been walking for four million years.*

How much did we once walk every day? Loren Cordain, a profes

*To put our history of mobility into perspective, try to picture time since the da the first hominid stood tall as a walk across New York's Central Park, all fifty on blocks from Harlem to Midtown Manhattan. We'd be hunter-gatherers from en to end for thousands of steps, right until the moment we could spot the doorman o the Plaza Hotel on Fifty-ninth Street. The age of farming would almost add up to th sprint across Fifty-ninth. We'd enter the age of cities on the sidewalk right in fron of the hotel. The years during which we've let automobiles do the work for us woul take up less than the depth of one red carpet–clad step at the hotel's front door

sor of health and exercise science at Colorado State University, tried to find out by comparing the daily energy expenditure of the average sedentary office worker to modern hunter-gatherers such as the !Kung of southern Africa. In some parts, !Kung women still spend their days collecting nuts, berries, and roots while the men hunt lizards, wildebeests, and whatever else they can track down in the desert. The women tend to walk about six miles per day and the men as much as nine, often burdened by heavy loads. The average American office worker gets barely a fifth of that exercise.

This is a troubling state of affairs, given that immobility is to the human body what rust is to the classic car. Stop moving long enough, and your muscles will atrophy. Bones will weaken. Blood will clot. You will find it harder to concentrate and solve problems. Immobility is not merely a state closer to death: it hastens it. Just spending too much time sitting shortens your life span.

We have evolved to get smarter and cheerier when we exercise, provided we can do it someplace where we aren't burning, freezing, terrified, or in other mortal danger. Robert Thayer, a professor of psychology at California State University, fitted dozens of students with pedometers, then sent them back to their regular lives. Over the course of twenty days, the volunteers answered survey questions about their moods, attitudes, diet, and happiness. The average student walked 9,217 steps a day—much more than the typical American, though much less than a !Kung tribesman.* But within that volunteer group, people who walked more tended to feel more energetic and upbeat. They had higher self-esteem. They were happier. They even felt that their food was better for them.

"We're talking about a wider phenomenon here than just walk more, feel more energy. We're talking about walk more, be happier, have higher self-esteem, be more into your diet and also the nutritiousness of your diet," Thayer said. The psychologist has devoted his life to the study of human moods. In test after test he proved that the most powerful way to fix a dark mood is simply to take a brisk walk. "Walking works like a drug, and it starts working even after a few steps."

As the philosopher Søren Kierkegaard put it, there is no thought

*The average walking step is 2.5 feet. So the average student walked around 4.5 miles each day, much more than the typical American.

so burdensome that you cannot walk away from it. We can literally walk ourselves into a state of well-being.

The same is true of cycling, although a bicycle has the added benefit of giving even a lazy rider the ability to travel three or four times faster than someone walking, while using less than a quarter of the energy. A bicycle can expand the self-propelled travelers' geographical reach by an astounding nine or sixteen times. Quite simply, a human on a bicycle is the most efficient traveler among all machines and animals.

Even those who endure the most severe bicycle trips seem to take pleasure in them. They feel capable. They feel free. They feel and are healthier. The average convert to bike commuting loses thirteen pounds in the first year. They may not all attain Robert Judge's level of transcendence, but cyclists report feeling connected to the world around them in a way that is simply not possible in the sealed environment of an automobile or a bus or a subway car. Their journeys are both sensual and kinesthetic.

All this points to two problems in urban mobility. First, people are not maximizing happiness on their commutes, especially in North American cities. Second, and perhaps more urgent, most of us are overwhelmingly choosing the most polluting, expensive, and place-destroying way of moving. As I discussed in the previous chapter, cars, whether they are caught in congestion or moving fast and free, can rip apart the social fabric of neighborhoods. They are by far the biggest source of smog in most cities. They produce more greenhouse gas emissions per passenger mile than almost any other way of traveling, including flying by jet airliner. It seems preposterous that we would choose a way of moving that simultaneously fails to maximize pleasure while maximizing harm. But once again, we are not all as free to choose as we might hope.

Behavior by Design

Of every one hundred American commuters, five take public transit, three walk, and only one rides a bicycle to work or school. If walking and cycling are so pleasurable, why don't more people choose to cycle or walk to work? Why do most people fail to walk even the ten thousand daily steps needed to stay healthy? Why do we avoid public transit?

I was naive enough to ask that question of a fellow diner I met in the food court of the bunkerlike Peachtree Center in downtown Atlanta. Her name was Lucy. She had driven her car in that morning from Clayton County (a freeway journey of about fifteen miles), pulled into a parking deck, followed a skyway a few dozen paces to an elevator and then a few more to her desk. Trip time: about half an hour. Total footsteps: maybe three hundred. She flashed me a broad smile.

"Honey, we don't *walk* in Atlanta," Lucy told me. "We *all* drive here. I can't say why. I guess we're just lazy."

Lazy? The theory doesn't stand up. Lucy's own commute was proof. She could not have made it to work any other way. Suburban Clayton County parked its entire bus fleet in 2010.* In the midst of a cash crunch, the county just couldn't afford to run buses through the sparsely populated dispersed city.

No, the answer to the mobility conundrum lies in the intersection between psychology and design. We are pushed and pulled according to the systems in which we find ourselves, and certain geometries ensure that none of us are as free as we might think.

Few places design travel behavior as powerfully as Atlanta. The average working adult in Atlanta's suburbs now drives forty-four miles a day.† Ninety-four percent of Atlantans commute by car. They spend more on gas than anyone else in the country. In Chapter 5, I explained how the centrifugal force of Atlanta's extensive freeway network enabled its population to spread far across the Georgia countryside—and then left them vulnerable to world-class road congestion. But in a study of more than eight thousand households, investigators from the Georgia Institute of Technology led by Lawrence Frank discovered that people's environments were shaping their travel behavior and their bodies. They could actually predict how fat people were by where they lived in the city.

Frank found that a white male living in Midtown, a lively district near Atlanta's downtown, was likely to weigh ten pounds less than

* The bus service in Clayton County carried two million riders in 2009 before it was shut down.

† That's seventy-two minutes a day behind the wheel, just getting to work and back.